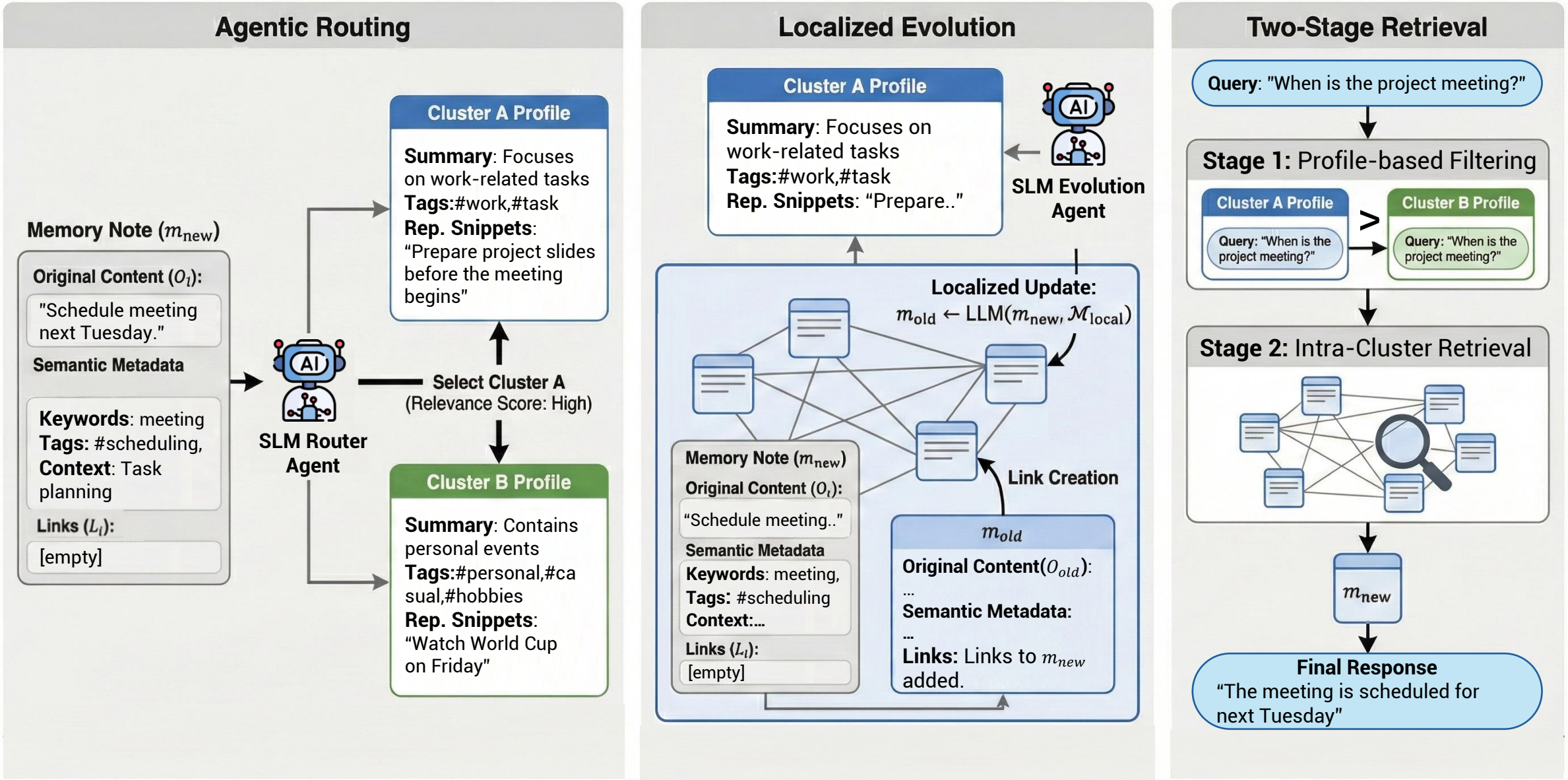
Agentic Routing
Memory Note (m_{new})
Original Content (O_i):
"Schedule meeting next Tuesday."
Semantic Metadata
Keywords: meeting
Tags: #scheduling,
Context: Task planning
Links (L_i):
[empty]
SLM Router Agent
Select Cluster A
(Relevance Score: High)
Cluster A Profile
Summary: Focuses on work-related tasks
Tags:#work,#task
Rep. Snippets: "Prepare project slides before the meeting begins"
Cluster B Profile
Summary: Contains personal events
Tags:#personal,#casual,#hobbies
Rep. Snippets: "Watch World Cup on Friday"
Localized Evolution
Cluster A Profile
Summary: Focuses on work-related tasks
Tags:#work,#task
Rep. Snippets: "Prepare.."
SLM Evolution Agent
Localized Update:
$m_{old} \leftarrow \text{LLM}(m_{new}, \mathcal{M}_{local})$
Memory Note (m_{new})
Original Content (O_i):
"Schedule meeting.."
Semantic Metadata
Keywords: meeting,
Tags: #scheduling
Context:...
Links (L_i):
[empty]
Link Creation
m_{old}
Original Content(O_{old}):
...
Semantic Metadata:
...
Links: Links to m_{new} added.
Two-Stage Retrieval
Query: "When is the project meeting?"
Stage 1: Profile-based Filtering
Cluster A Profile
Query: "When is the project meeting?"
>
Cluster B Profile
Query: "When is the project meeting?"
Stage 2: Intra-Cluster Retrieval
m_{new}
Final Response
"The meeting is scheduled for next Tuesday"